Chaitanya Priya

Senior Embedded Software Engineer | EV Charging, Diagnostics & FuSi

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PROFESSIONAL SUMMARY

Embedded software specialist with over 10 years of hands-on experience in safety-critical automotive systems, with strong focus on **charging control software**, **diagnostics**, and robust **fault handling**. Proven delivery of control functions for **charging modes** and communication paths, diagnostic implementations (**OBD/UDS/CAN**), and **FuSi-compliant** software integration according to **ISO 26262**. Experienced across **AUTOSAR** architecture and implementation (BSW/MCAL), safety concepts (**watchdog**, self-test, plausibility checks), and end-to-end verification through **unit**, **integration**, **SIL/HIL**, and **system tests**. Certified in Automotive Cybersecurity (ISO 21434, TÜV SUD) and Functional Safety (ISO 26262, Omnex), with ASPICE-compliant delivery background at Mercedes-Benz, Vector Informatik, American Axle & Manufacturing, and John Deere.

CORE COMPETENCIES

Programming	Embedded C, C++, Python, Structured Text
Charging & Diagnostics	**Charging modes** and communication control, **UDS** (ISO 14229), **CAN/OBD diagnostics**, **DTC handling**, diagnostic services and calibration support
Protocols & Interfaces	CAN, CAN FD, UDS (ISO 14229), SAE J1939, I2C, SPI, UART, MQTT, ModBus RTU/ASCII, PROFINET, RS-485
Microcontrollers & RTOS	Aurix TCx, STM32, Renesas (16/32-bit), C8051F, HSM/SBC integration, FreeRTOS, JDOS, RTA-OS, Bare-Metal
Automotive Standards	**AUTOSAR** (BSW, MCAL, DaVinci Configurator/Developer, ETAS ISOLAR-A/B, RTA-BSW), **ISO 26262**, ISO 21434, ASPICE (SWE.1–SWE.5), **FMEDA** support
Tools & Test	MATLAB & Simulink, Embedded Coder, **Vector CANoe**/CANalyzer/CANape, **DaVinci Toolchain**, **EB tresos**/MCAL, VectorCast, vTESTstudio, Polyspace BF/CP, Lauterbach TRACE32, VFlash, ASAP2
Methodology & DevOps	**SIL**, **HIL**, MIL, CI/CD, MISRA, **watchdog**/self-test/plausibility checks, 8D, FMEA, DFMEA, OpenOCD, JTAG
Design & PLM	KiCAD, Altium Designer, PreeVision, IBM DOORS NG, IBM RTC, GIT, JIRA, Confluence
AI & Data	RAG Pipelines, LLM-driven Root Cause Analysis, Generative AI (IIT Certified)

PROFESSIONAL EXPERIENCE

Senior Technical Lead

Oct 2024 – Present

Mercedes-Benz GmbH (via Soorce Professional GmbH) — Stuttgart, Germany

- Developed and maintained the Power State Manager (PSM) for Aurix TCx-based boards and multi-OS SoCs (AUTOSAR, QNX, Linux via hypervisor)
- Implemented **control software** behavior for **charging-related** power states and communication handshakes across multi-OS domains
- Engineered multi-stage hardware initialization logic for seamless control handoff between Aurix, QNX, and Linux
- Integrated safety mechanisms including **watchdog** supervision, plausibility checks, and deterministic fallback behavior for **fault cases**
- Architected an automated Jira triage pipeline using RAG to classify and redirect out-of-scope tickets, reducing manual triage effort
- Implemented AI-driven root cause analysis to identify software defects and generate targeted code fix suggestions
- Collaborated with the Architecture Team to refine the safety-critical “OnOff” system for ASIL-relevant power states
- Resolved complex hardware-software interface defects by owning the end-to-end bug resolution lifecycle

- Supported **software architecture** and technical documentation for ASPICE/FuSi work products and release readiness
- Delivered work products aligned with ASPICE SWE.1 through SWE.4

Senior Technical Lead

Jul 2023 – Sep 2024

Vector Informatik GmbH (via MSK Engineering and IT Services GmbH) — Stuttgart, Germany

- Led requirements elicitation and developed comprehensive design documentation for EV charging sequences
- Engineered complex device drivers and designed robust low-level software architectures
- Designed software solutions and integrated third-party libraries into existing architectural frameworks
- Configured and maintained AUTOSAR base software to ensure optimal system performance
- Implemented **diagnostic** and **fault-handling** software functions for charging ECUs, including **UDS** services and **DTC** workflows
- Coordinated with global customers and vendors to facilitate integration of third-party libraries
- Developed rigorous test and safety plans to ensure full compliance with ISO 26262 standards
- Executed comprehensive **unit**, static code analysis, **SIL**, and **integration testing** as Feature Owner
- Executed **integration** and **system-level validation** in **CANoe**, including diagnostic communication and edge-case fault scenarios
- Represented the team in ASPICE Quality and ISO 21434 Cybersecurity Audits to ensure compliance

Lead Embedded Software Engineer

Oct 2019 – Jul 2023

American Axle & Manufacturing (AAM) — Detroit, USA

- Developed device drivers using Embedded C for Motor Control ECU integration with RTA-OS on Aurix platform
- Created robust test plans and performed SIL/HIL testing to reduce bugs in production
- Developed Embedded C/C++ software for vehicle diagnostics in accordance with ISO 14229 (UDS)
- Integrated third-party libraries including Infineon SafeTlib and Cycurlib for security and safety compliance
- Configured AUTOSAR Base Software (BSW) and Microcontroller Abstraction Layer (MCAL)
- Executed unit testing, static code analysis (Polyspace), and SIL/integration testing as Feature Owner
- Managed board bring-up, driver development, and BSP configuration for POCs and new platforms
- Coached team members on technical blockers to ensure successful achievement of delivery milestones
- Participated in customer and internal ASPICE audits (SWE.1–SWE.5) and defined corrective actions

Lead Engineer

Sep 2017 – Oct 2019

John Deere (via HCL Technologies) — Pune, India

- Developed device drivers for SPI- and I2C-based CAN controllers and external memory modules
- Migrated application layer features from 16-bit microcontrollers to the 32-bit Renesas platform
- Tested features using Model-in-the-Loop (MIL) and Software-in-the-Loop (SIL) methodologies
- Generated and integrated production C/C++ code using Embedded Coder for model-based development
- Maintained and developed CI/CD pipelines (Vehicle Payload Delivery) for smooth deployments
- Integrated, built, and debugged code for JDOS (John Deere RTOS) based vehicle projects
- Resolved defects and performed root cause analysis using 8D, FMEA, and DFMEA techniques
- Conducted reviews of specifications, code, and quality reports to recommend improvements
- Evaluated technical feasibility and commercial impact of change requests with offshore management

Embedded Systems Software Developer

Feb 2015 – Sep 2017

Libratherm Instruments Pvt. Ltd. — Mumbai, India

- Developed and maintained firmware for medical devices, ensuring compliance with FDA regulatory standards
- Programmed applications and modules for STM32 microcontrollers using C/C++
- Integrated communication technologies and IoT protocols including MQTT and ModBus RTU/ASCII
- Designed firmware for multi-zone PID temperature controllers (6-zone ramp/soak profiles, RS-485 Master/Slave)
- Developed control software for industrial systems: precision thermal conditioning, helium leak detection (PLC, Structured Text)
- Created comprehensive documentation required for product approval and certification processes
- Led the full product lifecycle from initial concept through mass production and aftersales support
- Conducted rigorous unit tests to ensure firmware reliability and system performance

Trainer - Embedded Systems Software Aug 2012 – Jan 2015

Gencor Learning Solutions Pvt. Ltd. — Pune, India

- Created structured lessons on microcontroller architectures, memory maps, and register-level programming
- Conducted live sessions on "board bring-up," where trainees learn to initialize hardware from scratch
- Guided students in writing drivers for GPIO, UART, SPI, I2C, ADC, and PWM without high-level libraries
- Prepared trainees by teaching best practices in version control (Git), MISRA C standards, and RTOS like FreeRTOS
- Constantly monitored trainee progress by devising new tests and lab exercises

EDUCATION

Embedded Systems Design	University of Pune, India
B.Sc. Engineering (Distinction)	Magadh University, India

CERTIFICATIONS

Certified in ML and Generative AI for Professionals	Indian Institute of Technology (IIT)
ISO 21434:2021 – Automotive Cybersecurity Practitioner	TÜV SÜD
ISO 26262:2018 – Functional Safety Level 1	Omnex

LANGUAGES

English	Fluent (C2)
German	Professional Working (B2, ongoing improvement)
Hindi	Native